Oxygen and water

Of course, life in space would be impossible without the two most important factors for humans to stay alive – oxygen and water.

Life support systems on the ISS must not only supply oxygen and remove carbon dioxide from the cabin's atmosphere, but also prevent gases like ammonia and acetone, which people emit in small quantities, from accumulating. Vaporous chemicals from science experiments are a potential hazard, too, if they combine in unforeseen ways with other elements in the air supply.

Most of the station's oxygen comes from a process called electrolysis, which uses electricity from the ISS solar panels to split water into hydrogen gas and oxygen gas. Each molecule of water contains two hydrogen atoms and one oxygen atom. Running a current through water causes these atoms to separate and recombine as gaseous hydrogen (H2) and oxygen (O2).

Carbon dioxide is removed from the air by a machine based on a material called zeolite, which acts as a molecular sieve, according to Jim Knox, a carbon dioxide control specialist at MSFC. The removed CO2 is vented to space. Engineers are also thinking of ways to recycle the gas.

In addition to exhaled CO2, people also emit small amounts of other gases. Methane and carbon dioxide are produced in the intestines, and ammonia is created by the breakdown of urea in sweat. People also emit acetone, methyl alcohol and carbon monoxide – which are byproducts of metabolism – in their urine and their breath. Activated charcoal filters are the primary method for removing these chemicals from the air.

The Water Recycling System (WRS) reclaims waste waters from the fuel cells, from urine, from oral hygiene and hand washing, and by condensing humidity from the air. Without such careful recycling, over 18,000 kilograms of water from Earth would be required per year to resupply a minimum of four crewmembers for the life of the station.

It might sound disgusting, but water leaving the space station's purification machines is cleaner than what most of us drink on Earth. It goes through an aggressive treatment process, giving practically ultra-pure water in the end.

The first step is a filter that removes particles and debris. Then the water passes through the "multi-filtration beds", which contain substances that remove organic and inorganic impurities. And finally, the catalytic oxidation reactor removes volatile organic compounds and kills bacteria and viruses.

However, even with intense conservation and recycling efforts, the ISS